CS 6120: Comparing Models for Multilabel Classification of PubMed Article Abstracts

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Abstract

In this paper, we discuss our project on multi-label classification of PubMed articles into root labels using NLP. We approach the biomedical classification task with BERT (BioBERT) and compare it against three baseline models: RNN, CNN, and Logistic Regression.

1 Introduction

The growth of biomedical literature has created a need for automated methods to organize and classify research articles effectively. Manually labeling articles with relevant Medical Subject Headings (MeSH) is labor-intensive and time-consuming, making automated multi-label classification a helpful tool for researchers and clinicians.

This project aims to use a transformer-based language model, specifically BERT, to perform multi-label classification on biomedical research articles. Instead of traditional single-label classification, we use multi-label classification, as articles are typically associated with multiple MeSH tags.

Our goals are: (1) to train and evaluate a BERT model for accurately predicting MeSH root labels and (2) to compare its performance against more conventional models, including logistic regression, CNNs, and RNNs. By doing so, we hope to demonstrate the advantages of transformer-based models in capturing contextual information and label correlations in biomedical texts, ultimately providing a tool for organizing and analyzing the PubMed corpus.

In our classification task, we encounter challenges such as fixing class imbalance, dealing with specialized language, and optimizing training time / compute. Different evaluation metrics and source data resizing became part of the solution to some of these problems. Through this report, we detail our data processing, model architectures, experimental setup, and evaluation results, along with an analysis of each approach.

2 Background and Related Work

The rapid rise in biomedical publications makes it hard for scientists to find useful data. To help, the National Library of Medicine (NLM) tags PubMed entries using Medical Subject Headings (MeSH), a structured list grouping medical ideas in levels. Yet, tagging by hand takes much effort, money, and time - making expansion tough (Koutsomitropoulos and Andriopoulos, 2020). For this reason, automatic Multi-Label Text Classification (MLTC) has become vital in bio-NLP tasks. In contrast to typical single-label sorting, MLTC links several tags to one document, since most health papers cover various subjects at once - for example Disorders, Substances, or Treatment Methods (Tsoumakas and Katakis, 2009).

Before Transformers became common, models for text analysis often used CNNs instead of RNNs. While both handled sequences differently, they formed the base for early deep learning in this area. CNNs started in image processing but later got used for text tasks after Kim's work (Kim, 2014), using small sliding filters to spot short meaningful word groups. Although good at finding fixed-length phrases, they usually fail to link distant parts of longer texts because their focus area is narrow. RNNs plus Bi-LSTMs: recurrent systems like Bidirectional Long Short-Term Memory (Bi-LSTM) emerged to handle text step by step. Because they analyze data from left to right - then reverse - they grasp sequence patterns along with word positioning better than fixed bag-of-words methods (Hochreiter and Schmidhuber, 1997). Still, such architectures run into processing delays thanks to their linear flow; also, gradients tend to weaken across extended chains.

Devlin et al. (Devlin et al., 2019) introduced BERT along with this new structure, changing how NLP works. Instead of using RNNs, these models rely on self-attention to handle input at once -

enabling better understanding of context and links between labels.

Although standard BERT works well on common tasks, it struggles with biomedical texts because of differences in terminology - like drug names or clinical terms. Instead of using general data, Lee et al. (Lee et al., 2019) trained BioBERT on specialized sources such as PubMed and PMC, keeping the original structure. This version performs better in spotting entities, finding links between them, also answering medical questions. Here, we use biobert-base-cased-v1.2 to identify specific patterns tied to MeSH labels.

3 Data

The baseline and BERT models were all trained and tested on the dataset PubMed MultiLabel Text Classification Dataset MeSH. This dataset contains information on 50,000 articles from PubMed, including each article's title, abstract, and Medical Subject Heading (MeSH) labels.

MeSH labels are arranged into a nested hierarchy with 16 root labels (see Table 1). Two labels—Humanities [K] and Publication Characteristics [V]—have been excluded due to their infrequency in the available data. The 14 included labels each appear in at least ten percent of the sample articles (see Figure 1).

Each article has up to 13 root labels, with a median root label count of six (see Figure 2). The article abstracts are an average of 222 tokens long (see Figure 3).

4 Methods

We adjusted the BioBERT model (biobert-base-cased-v1.2) to assign medical summaries to up to 14 MeSH groups that may overlap. Texts were processed using BioBERT's cased WordPiece tokenizer, which inserts special [CLS] and [SEP] tokens and pads or truncates sequences to a fixed length of 128 tokens. Instead of stacking multiple layers, a single linear module was connected to the model's final output, transforming hidden states into 14 separate prediction values. This meant the final hidden state of just the [CLS] token was passed through a single linear layer that produced 14 independent logits. Since articles might fit more than one label at once, outputs used individual sigmoid functions per node, avoiding softmax which assumes exclusive classes. This setup allowed parallel probability estimates across topics. Training relied on BCEWithLogitsLoss—a unified method combining sigmoids and binary loss—for better computational reliability during optimization. The entire BioBERT encoder and classification head were fine-tuned jointly using the AdamW optimizer. At prediction time, a fixed cutoff of 0.5 was used; categories surpassing the cutoff were linked to the text.

The performance of the fine-tuned BERT model was measured against three baseline models which were trained as follows.

The logistic regression model serves as a basic neural starting point while performing effectively like a "Neural Bag-of-Words" system. Although it uses an embedding layer set to 256 dimensions by default, this component converts word IDs into compact vector forms. Rather than tracking sequence position, it computes the average of these embedded values throughout the full text entry. As a result, one consolidated mean vector emerges, capturing the overall meaning of the abstract. This average form feeds straight into a linear layer, then uses BCEWithLogitsLoss for training. The approach estimates each label's likelihood separately, yet maintains fast response times.

The CNN approach detects local word sequences and meaningful phrases in text, no matter where they appear. Instead of focusing on exact positions, it processes embedded inputs with length 256 using a 1D convolutional layer with a window size of 50 and moves forward by steps of 20 to capture distinct signals from the abstract. Following this step, global max pooling selects the strongest activation per filter over the full input span. Then, these selected outputs use BCEWithLogitsLoss for training so that multiple labels can be predicted at once.

The RNN model handles word order by applying a Bi-LSTM setup. First, an embedding step converts split text into 256 element vectors. Then, these representations enter a Bi-LSTM block containing 128 memory cells. Instead of one direction only, processing occurs left-to-right and right-to-left to grasp full-sentence meaning. Afterward, outputs from both flows merge before entering a dense layer. That last component computes unnormalized scores directly. During inference, logits become probabilities using a sigmoid activation. With a set cutoff, multi-label outputs turn into binary results.

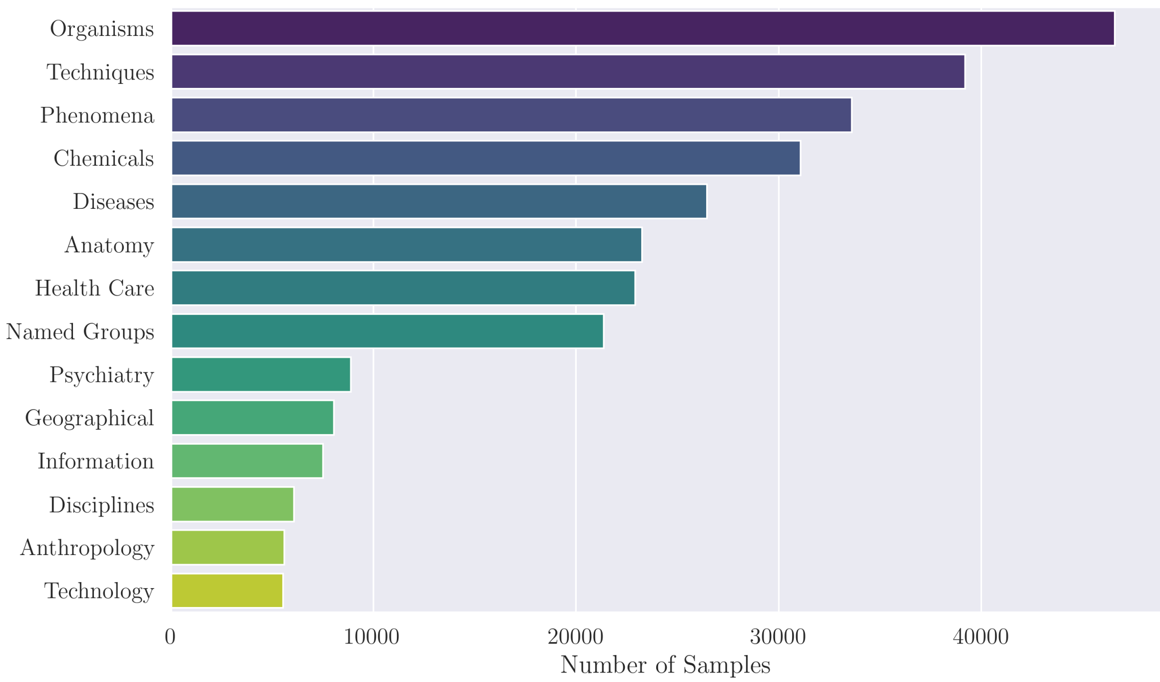

Figure 1: Frequency of MeSH Root Labels

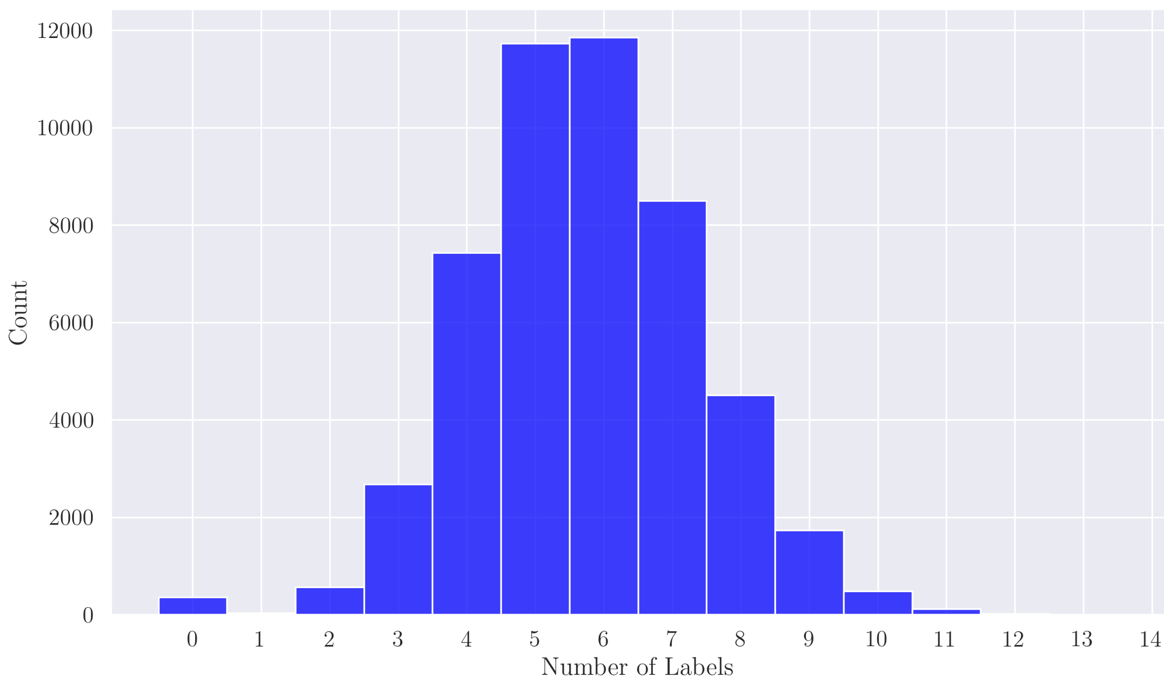

Figure 2: Number of Labels per Abstract

5 Experiments

All of the baseline models were trained over 25 epochs, using an initial learning rate of 0.001 with Adam optimization, on both reduced-size and full-size versions of the dataset.

At the end of each epoch, the micro F1, macro F1, weighted-average F1, exact match ratio, and Hamming loss were calculated. Micro F1 calculates the F1 score using precision and recall over all of the labels at once, while the macro F1 calculates the F1 for each label and returns the average

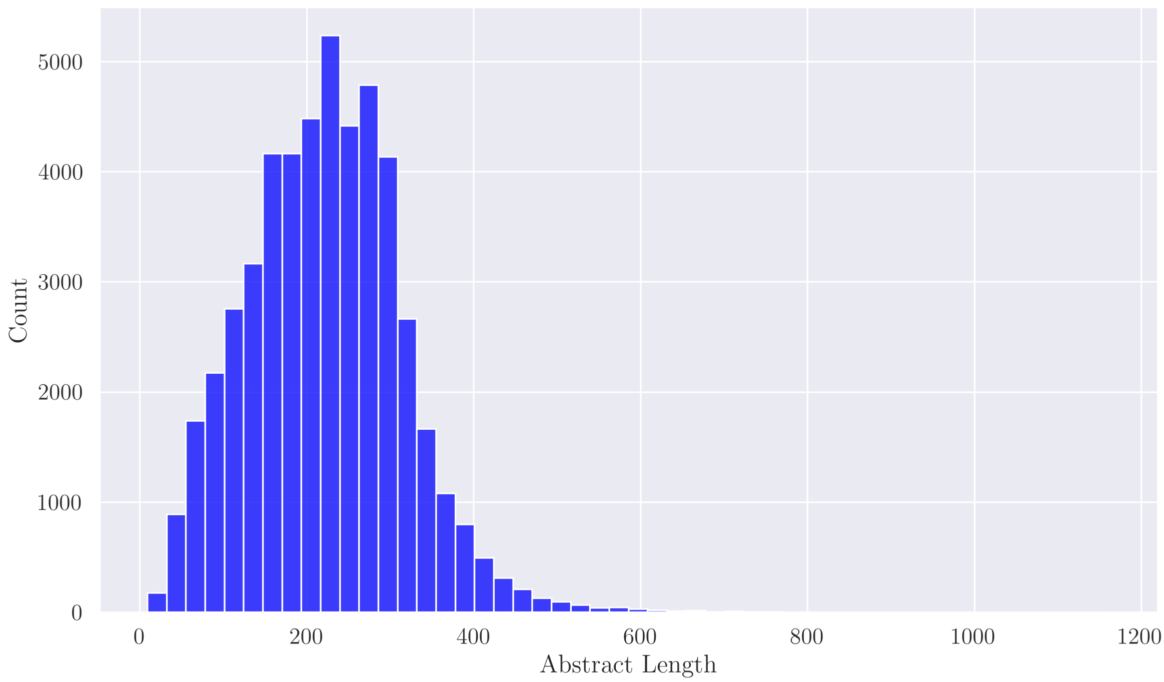

Figure 3: Distribution of Abstract Lengths

across the labels. Similarly to macro F1, weighted-average F1 calculates the F1 for each label and returns the weighted average. The exact match ratio measures how many predictions exactly matched every label, and Hamming loss measures how many labels were predicted incorrectly across all labels. Higher F1 scores and exact match ratios indicate better performance, and lower Hamming loss indicates better performance. Together, these five metrics were used to measure the performance of each of the models.

5.1 Results

Looking at the results, the BioBERT model leads baseline models in performance on the classification task. Micro and Macro F1 are higher for BioBERT compared to the baseline models, which shows that even with class imbalance BioBERT beats out the baseline. The amount by which BioBERT beats all three baselines is more for the smaller dataset, with performance at around 0.7-0.8, and this shows the difference between the BERT fine-tuning vs. training from start in the baseline models. This difference across reduced and full data continues across to Exact match ratio and hamming loss: BioBERT has lower hamming loss and higher exact match ratio than baseline.

The RNN model performed best of the three baseline models. It had the highest macro and weighted-average F1 scores of the three baseline models over all 25 epochs, and performed best with regard to micro F1, exact match ratio, and Hamming loss for most of training.

Although linear regression had worse performance than RNN across all five metrics for the first several epochs of training, the model using linear regression had the marginally better performance with regard to micro F1, exact match ratio, and Hamming loss past 15 epochs of training.

The CNN model had by far the worst performance, with lower F1 scores, lower exact match ratios, and higher Hamming loss for the majority of training. CNN only performed better than logistic regression for the first two epochs, and never performed better than RNN for any evaluation metric.

6 Conclusions

Present your discussion/conclusions.

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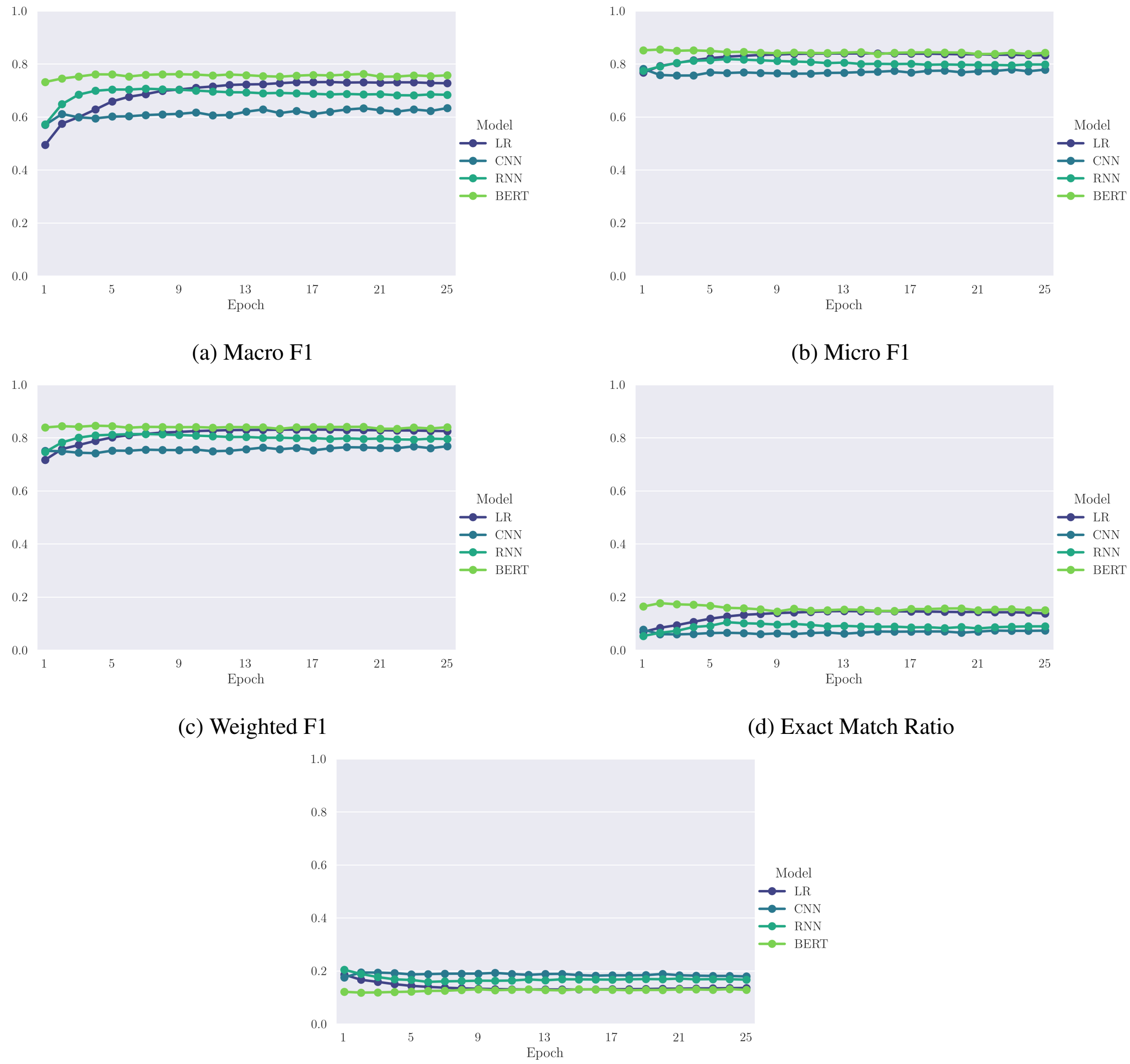

(a) Macro F1

(b) Micro F1

(c) Weighted F1

(d) Exact Match Ratio

(e) Hamming Loss

Figure 5: Performance Metrics by Epoch

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Label	Name
A	Anatomy
B	Organisms
C	Diseases
D	Chemicals and Drugs
E	Analytical, Diagnostic and Therapeutic Techniques, and Equipment
F	Psychiatry and Psychology
G	Phenomena and Processes
H	Disciplines and Occupations
I	Anthropology, Education, Sociology, and Social Phenomena
J	Technology, Industry, and Agriculture
K	Humanities
L	Information Science
M	Named Groups
N	Health Care
V	Publication Characteristics
Z	Geographicals

Table 1: Full Label Names